1. Suppose $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Compute e^A. (Hint: use the formula $e^{At} = I + At + \frac{A^2t^2}{2!} + \cdots \frac{A^nt^n}{n!} + \cdots$)

Gavin Kelley
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by hint provided, $e^{At} = I + A + \frac{A^2}{2!} + \dots + \frac{A^n}{n!} + \dots = \begin{bmatrix} a_{11} & \cdots & \\ \vdots & \ddots & \\ \vdots & & a_{33} \end{bmatrix}$ (where $t = 1 \ \forall n$)

<u>for this specific A matrix:</u>

$$A^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow A^3 = AA^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\boxed{A^i = 0 \quad \forall \quad i > 2} \quad (i \in \mathbb{N})$$

$$\Rightarrow e^A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{I} + \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}}_{A} + \underbrace{\frac{1}{2}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\frac{1}{2!}A^2} + (0) = \boxed{\begin{pmatrix} 1 & 1 & 1/2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}$$

3. Suppose $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Compute e^{At}.

Useful Formula:

- $e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots$

Represent A by Pauli matrix, $\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

(x-spin state represented in z-basis)

$\sigma_1^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$ (Involutory, & therefore Orthogonal)

$\sigma_1^3 = I\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \sigma_1$

$$\boxed{\text{for } n \in \mathbb{N}, \quad \sigma_1^{2n-1} = \sigma_1 \quad \& \quad \sigma_1^{2n} = I}$$

Now, $$e^{\sigma_1 t} = \sum_{n=0}^{\infty} \frac{\sigma_1^n t^n}{n!}$$

$$= \underbrace{\left(\sum_{n=0}^{\infty} \frac{\sigma_1^{2n} t^{2n}}{(2n)!}\right)}_{\text{even-powered terms}} + \underbrace{\left(\sum_{n=0}^{\infty} \frac{\sigma_1^{2n+1} t^{2n+1}}{(2n+1)!}\right)}_{\text{odd-powered terms}}$$

$$= (I)\sum_{n=0}^{\infty} \frac{t^{2n}}{(2n)!} + \sigma_1 \sum_{n=0}^{\infty} \frac{t^{2n+1}}{(2n+1)!}$$

$$= (I)\left(\frac{e^t + e^{-t}}{2}\right) + \sigma_1\left(\frac{e^t - e^{-t}}{2}\right)$$

$$= \begin{pmatrix} \frac{e^t + e^{-t}}{2} & \frac{e^t - e^{-t}}{2} \\ \frac{e^t - e^{-t}}{2} & \frac{e^t + e^{-t}}{2} \end{pmatrix}$$

- $(e^t + e^{-t}) = 2(1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \cdots + \frac{t^{2n}}{(2n)!} + \cdots)$

  and $(e^t - e^{-t}) = 2(t + \frac{t^3}{3!} + \frac{t^5}{5!} + \cdots + \frac{t^{2n+1}}{(2n+1)!} + \cdots)$

$$\therefore \boxed{e^{At} = \begin{pmatrix} \cosh(t) & \sinh(t) \\ \sinh(t) & \cosh(t) \end{pmatrix}}$$